Q1. Display a Message

Write an 8086 assembly program to display `HELLO WORLD` on the screen.

Q2. Display Your Name

Write a program to display your name, department, and semester on separate lines.

Q3. Add Two Numbers

Write a program to add two single-digit numbers and display their sum.

Q4. Subtract Two Numbers

Write a program to subtract one single-digit number from another and display the result.

Q5. Increment a Number

Write a program to store a number in a register, increment it by one using `INC`, and display the result.

Q6. Decrement a Number

Write a program to initialize a register with a number, decrement it using `DEC`, and observe the final value.

Q7. Move Data Between Registers

Write a program to move a value into AX, transfer it to BX, and display the register values using a debugger.

Q8. Add Three Numbers

Write a program to add three single-digit numbers and display their sum.

Level 2: Moderate (Questions 9 – 17)

Conditional jumps, loops, comparisons, and arithmetic

Q9. Check Even or Odd

Write a program to check whether a single-digit number is even or odd.

Q10. Find the Greater Number

Write a program to compare two numbers and display the greater number.

Q11. Find the Smaller Number

Write a program to compare two numbers and display the smaller number.

Q12. Positive or Negative

Write a program to determine whether a signed number is positive, negative, or zero.

Q13. Multiplication of Two Numbers

Write a program to multiply two unsigned numbers using the `MUL` instruction.

Q14. Division of Two Numbers

Write a program to divide two unsigned numbers using `DIV` and display the quotient and remainder.

Q15. Print Numbers from 1 to 10

Write a program using a loop to display the digits from 1 through 9, followed by 0 only if you handle its display separately.

Q16. Sum of Numbers from 1 to 10

Write a program using a loop to calculate the sum of integers from 1 to 10.

Q17. Factorial of a Number

Write a program to calculate the factorial of a small positive integer, such as 5, using a loop.

Level 3: Hard (Questions 18 – 25)

Arrays, procedures, nested loops, and string operations

Q18. Find the Largest Element in an Array

Write a program to find the largest number in an array containing five unsigned bytes.

Q19. Find the Smallest Element in an Array

Write a program to find the smallest element in an array of five numbers.

Q20. Calculate the Sum of an Array

Write a program to calculate the sum of all elements in an array of five unsigned bytes.

Q21. Count Even and Odd Numbers

Write a program to count how many even and odd numbers are present in an array of ten unsigned bytes.

Q22. Reverse an Array

Write a program to reverse the order of five elements in an array.

Q23. Check Whether a Character Is a Vowel

Write a program to accept a character from the keyboard and determine whether it is a vowel or consonant.

Q24. String Length

Write a program to calculate the length of a `$`-terminated string without counting the terminating `$` character.

Q25. Sort an Array in Ascending Order

Write a program to sort five unsigned numbers in ascending order using Bubble Sort and nested loops.